Tandem: Master Design Document

October 7, 2026

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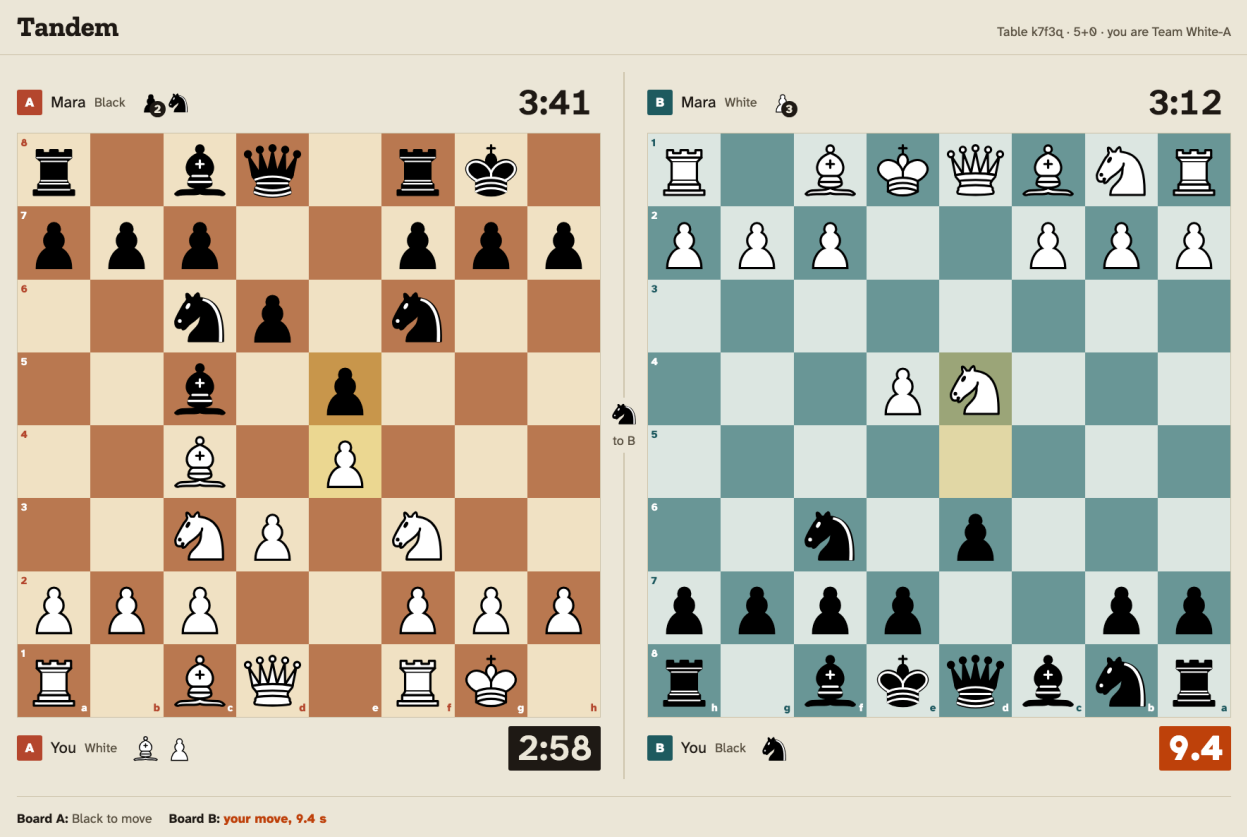
Overview

Tandem is an online two-player bughouse chess site where each player controls both boards, with one clock per board.

Compiled 2026-10-07 from the project files. This is a read-only compilation; edit the source files (`BRIEF.md` , `DESIGN.md` , `THIRD_PARTY.md` , `design/tokens.css`), then rebuild with `python3 tools/build_master.py` .

Status at a glance

Area	State
Brief	v2.1, complete, no open questions
Visual direction	**Club table**, decided 2026-10-07
Tokens, fonts, pieces, licences	Done and verified (`design/` , `THIRD_PARTY.md`)
Rules engine (M1)	Done. Run for this compilation: **49 tests passing** (3 files)
Local prototype (M2)	Not started; next
Server, UI, replay, hardening (M3 to M6)	Not started
Runtime dependencies	None yet; each is licence-checked when added



Part 1. Build brief

Status: **v2.1, 2026-10-07.** Design direction decided (Club table). Rewritten after a line-by-line ambiguity audit and given a final consistency pass; see the changelog at the end. Working title: **Tandem** (placeholder, §14).

Reading guide: “must” is a requirement; “default” is a decision the user may reverse; every number in this document is a decision, not a suggestion.

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0. Summary

A website where two people play real-time bughouse chess online, and **each person controls both boards**. Player 1 plays White on Board A and Black on Board B. Player 2 plays Black on A and White on B. A player’s two seats are partners, so every piece captured on one board goes straight into *the same player’s* reserve on the other board. The game turns on attention, timing and piece flow instead of partner coordination.

There is **one clock per board.** Each is a normal two-sided chess clock, so each of the four seats has its own time bank. A player cannot ignore a board without paying for it in time. Rules follow established bughouse rules (§3). The design is the “Club table” direction (§7), chosen to avoid the AI-generic look and our earlier dark, data-dense projects.

0.1 GLOSSARY (WORDS ARE USED ONLY IN THESE SENSES)

Term	Meaning
Table	The pre-game room at `/t/:id`, before and during setup
Game	One played game from the start signal to a result
Board A / Board B	The two boards. A is always on the left, B on the right
Seat	One side of one board: `A–w`, `A–b`, `B–w`, `B–b`
Team	The two partner seats owned by one human: **Team White-A** = `A–w` + `B–b`; **Team Black-A** = `A–b` + `B–w`
Reserve	Pieces a seat may drop. Never “pocket”, “bank” or “hand” in the UI
Drop	Placing a reserve piece on an empty square as the seat’s action
Action	A move or a drop
Pending	In check, nothing legal now, but a dropped piece could block (§3.4)
Counterfeit	A promoted piece, which becomes a pawn when captured

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1. Decisions locked by the user

#	Decision	When
D1	Online 1v1, two devices, shared link, real-time server	2026-10-06
D2	Each human controls both boards (two partner seats)	2026-10-06
D3	**One clock per board** (classic bughouse clocks), replacing the earlier “one clock for both boards”, so players are forced to play both boards	2026-10-06
D4	Drops may give check and checkmate	2026-10-06
D5	All other rules follow the official/standard bughouse ruleset (§3)	2026-10-06
D6	Time control is chosen in the table by the host	2026-10-06
D7	Clean, organized UI. Not like previous projects. Not generic AI output	2026-10-06
D8	**Visual direction: “Club table”**	2026-10-07
D9	Fonts must not look like AI defaults (replaced, §7.4)	2026-10-07

Consequence of D2 + D3: the two boards run **concurrently in real time**, as in 4-player bughouse. This is not the fixed-rotation turn order of single-clock “simultaneous bughouse” (WhiteA → WhiteB → BlackB → BlackA). With one clock per board a rotation has no function, so it is not used. At any moment a player may have the move on both boards, one, or neither, and chooses where to spend attention. Each seat’s clock runs only while it is that seat’s move on its board.

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2. Research basis and rulings

Sources (links in §15): Wikipedia “Bughouse chess”, the wccc2007 “Laws of Bughouse” PDFs, the pion.ch complete ruleset, the chess.com Help Center bughouse article, the wikibooks Bug House page, and lichess/TalkChess discussions for the castling question. They agree on the core. There is **no FIDE-endorsed bughouse ruleset**, so sites and tournaments differ at the margins.

Confirmed by at least one source:

- Teams are partners across boards: White on A + Black on B against Black on A + White on B.
- A captured piece goes to the capturer’s partner immediately and keeps its own colour, which is the partner’s colour.
- A turn is a normal move **or** a drop on any empty square.
- Pawns cannot be dropped on rank 1 or rank 8. A pawn on its own second rank (rank 7 for Black), including one just dropped there, may double-step. A dropped pawn can be captured en passant.
- Promoted pieces revert to pawns when captured.
- A check that a dropped piece could block is **not** checkmate, even if the reserve is empty. Mate therefore needs a check that cannot be blocked by a drop: contact check, knight check, pawn check, double check, or a slider check with no empty square between.
- Stalemate does not exist: a side with no legal action waits.
- The game ends when either board ends (mate, resignation, flag).
- Drops may give check or mate in the standard rules (a shogi-style ban is a stated variant, rejected by D4).

Not settled by any authority, so decided here:

- **U1. Castling with a dropped rook.** Wikipedia: a rook dropped on a1/h1/a8/h8 may take part in castling. Lichess forbids it in *crazyhouse*, which is a different game, and discussions show disagreement. **Default: allowed**, requiring the king to be unmoved and the rook to have arrived on its corner by drop or never to have left it. It sits behind an engine flag `droppedRookCastling`, and the /rules page states the choice.
- **U2. Promotion choices.** OTB rulesets say “queen or knight” mainly because the pawn physically stays a pawn. **Default: Q, R, B or N.**
- **U3. Repetition and 50-move draws.** No standard definition exists and clocks always end the game. **Default: none** (§3.9).
- **U4. Piece set. Decided: Cburnett** (author Cburnett). All twelve SVGs were confirmed as **CC BY-SA 3.0** through the Wikimedia Commons API, so that is the licence we rely on (the author also offers GFDL, GPL and BSD on at least one file). Attribution is shown on /rules. Recorded in `THIRD_PARTY.md`. A custom set is a possible later upgrade, not v1.

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3. Rules specification (the engine contract)

3.1 SETUP

- Two standard boards, each in the standard starting position.
- Seats and partners: `A–w` ↔ `B–b` (Team White-A) and `A–b` ↔ `B–w` (Team Black-A).
- **Human 1** owns one team and **Human 2** the other (assignment in §4.1).
- Each seat’s time bank starts at the table’s base time.
- Colour never changes. A White piece captured on Board A (by `A–b`) goes to the reserve of `B–w`, still White. Every reserve therefore holds pieces of its seat’s own colour.

3.2 ACTIONS

An action is either:

1. **Move.** Standard chess movement, including castling, en passant and promotion.
2. **Drop.** Take a piece from the seat’s own reserve and place it on any empty square of the seat’s board.

Constraints:

- No action may leave your own king attacked. In check, only a move that resolves the check or a drop that blocks it is legal.
- **Pawn drops:** never on rank 1 or rank 8 (absolute ranks, both colours).
- **Double step:** a pawn that stands on its own second rank (rank 2 for White, rank 7 for Black) may advance two squares if both squares are empty. This includes a pawn just dropped there. A pawn dropped on any other rank advances one square.
- En passant works against dropped pawns exactly as against any pawn.
- Dropped pawns promote normally.
- Kings are never captured and never dropped.
- A seat may act only when it is that seat’s turn on its board.

3.3 CAPTURE TRANSFER

- A capture on board X by seat S adds the captured piece to the reserve of S’s partner seat on the other board. It becomes available at the instant the capturing action is applied, and the partner seat may use it **the next time that seat has the move**.
- **Promoted pieces** carry a `promoted` flag from promotion onward, kept as the piece moves. A captured promoted piece enters the reserve as a **pawn**.
- A capture that also gives check or mate still transfers its piece.
- En passant transfers a pawn.

3.4 CHECK, CHECKMATE AND PENDING

Applies to the seat to move, when its king is attacked. A **slider** is a bishop, rook or queen.

1. **Check:** at least one legal action exists with the reserve held now. Play continues.

2. **Checkmate:** no legal action exists now, **and** no drop could ever block. That means any of: more than one attacker (double check), an attacker that is a knight or pawn, or a slider attacker with no empty square between it and the king.
3. **Pending:** no legal action exists now, but exactly one slider attacks and at least one empty square lies between it and the king, so a dropped piece could block. The side is **not mated**. Its clock keeps running, and it may act as soon as its reserve (fed by its partner seat) gives it a legal block, or it may lose by flag or resignation. While pending, its opponent on that board cannot move, because it is not their turn.

A block square on rank 1 or rank 8 can only be filled by a non-pawn. Pending still applies, because a non-pawn could arrive.

Because a player's partner seat is also themselves, pending is rarely luck: they can usually produce the needed piece by capturing on their other board, if the clock allows.

3.5 NO STALEMATE

A side with no legal action that is **not** in check is called **stuck**. It is neither stalemated nor drawn. It waits with its clock running until a piece arrives or it flags. If neither seat on a board can act, that board is frozen until the other board changes a reserve or a clock expires.

3.6 END OF GAME

The game ends at the first of these events:

Event	Result
Checkmate on either board (§3.4 rule 2)	The team of the mated seat loses
Flag: a seat's time bank reaches 0 while its clock runs	The team of that seat loses. Flag is a loss even if the opponent could never mate; there is no insufficient-material rule
Resignation	The resigning human's team loses. It ends the whole game, not one board
Draw by agreement	Draw (§3.6.2)
Abandonment	§6.5
Abort	No result, §3.7.1

3.6.1 Order of events (deterministic)

- The server processes events in the order it receives them, one at a time.
- An action is accepted only if its server receive time (after lag credit, §6.2) is **at or before** the mover's flag deadline. An action received exactly at the deadline is accepted. Otherwise the flag event happens first and the action is discarded.
- Two game-ending events cannot come from two actions, because actions are serialized. They can only be two flags whose deadlines fall on the same millisecond. Then: if both seats belong to the **same team**, that team loses; if they belong to **different teams**, each team has lost one board, so the game is a **draw**.

3.6.2 Draw offers

- A human offers a draw; the other accepts or declines.
- An offer expires after 30 s, when the opponent declines, or when the game ends.
- At most one offer is pending. After a declined or expired offer, the same human may not offer again for 60 s.

3.7 CLOCKS

- Four time banks, one per seat. On each board only the seat to move has a running clock, so at most two clocks run at once (one per board), and none run on a frozen board.
- **Time control** (set by the host, default 5+0): base 1 to 30 minutes in whole minutes, applied to **each of the four seats**; increment 0 to 10 s in whole seconds. Presets: 1+0, 2+0, 3+0, 5+0, 10+0, 3+2, and custom. Standard bughouse has no increment; the default has none.

- **Increment:** Fischer. After each accepted action by a seat, add the increment to that seat's time bank, applied on the same board's clock. Not added on a flag.
- Time spent pending, stuck or deciding still runs.
- Display: at 10 s or more, `m:ss`; under 10 s, `s.t` (e.g. `9.4`). **Low time** is when a running clock is at or below `min(20 s, 10% of base)`.

3.7.1 Start and abort

- The game starts when both humans have pressed Ready, followed by a 3 s countdown. At the end of the countdown (`t0`), the White seat clocks on both boards start.
- If **neither** human has made any action 30 s after `t0`, the game is **aborted** with no result and no rematch penalty. Once any action has been made, only flags and the other end events in §3.6 apply.

3.8 ILLEGAL INPUT

The UI never offers illegal actions, and the server rejects them. There is no "illegal move loses" rule (that polices physical play). A rejected action reverts with a short message. A rejection never costs time beyond what elapsed.

3.9 DELIBERATELY ABSENT

No threefold repetition, no 50-move rule, no insufficient-material draw, no stalemate. Reserves and the other board change the state continually, no standard definition exists, and clocks always end the game. The /rules page says so.

3.10 NOTATION AND EXPORT

- Standard algebraic notation. Drops: `N@f3`, `P@e4`. Check `+`, mate `#`, promotion `e8=Q`, castling `0-0` / `0-0-0`.
- **Text export**, one line per action in order of server time: `<board>.<color> <notation> {<mover's remaining time after the action>}`, e.g. `A.w e4 {4:58.2}`, `B.b N@f3+ {3:41.0}`. A header of `Event`, `Date`, `Result`, `TimeControl`, `TeamWhiteA`, `TeamBlackA`.
- **JSON export:** the same list with `board`, `color`, `action`, `serverTs`, `clocks` after each action, plus the header and result.

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4. Product scope

4.1 IN V1

1. **Create a table.** The host chooses a time control and a team (Team White-A, Team Black-A, or random, resolved at the start of the game). The table has a link `/t/:id`.
2. **Join.** The first visitor who is not the host takes the other team. Anyone later is sent to the spectator view. A seat survives refresh through a seat token (§6.6). Guests enter a display name: 1 to 20 characters, letters, digits, space, `_` `-` `.`, trimmed; blank means `Guest` plus four digits; remembered locally.
3. **Table setup.** The host may change the time control or team until the guest has pressed Ready. Any change clears both Ready flags. An unjoined table expires after 30 minutes.
4. **Play.** Two live boards, four clocks, four reserves, drag-and-drop and click-click for moves and drops, keyboard entry, premoves, synthesized sounds, resign, draw offer.
5. **Result screen.** Winner and reason in one plain sentence, both final positions, the action list, and a rematch button. **Rematch:** either human can request it, the other has 60 s to accept, teams swap, settings are unchanged.
6. **Replay** at `/g/:id` (the same id; `/t/:id` forwards there when the game is finished): a timeline over the merged action list with step keys, both boards and all four reserves and clocks as recorded.
7. **Spectators** at `/watch/:id`: read-only, up to 50 concurrent, count shown to players. Orientation defaults to Team White-A's view, with a flip toggle.
8. **/rules:** the full ruleset with interactive mini-boards for each non-obvious rule (drop, counterfeit, pending, no stalemate, dropped-rook castling).
9. **Reconnect:** closing the tab or losing connection mid-game restores the same state on return (§6.4, §6.5).
10. **Export:** §3.10.

4.2 NOT IN V1

Accounts, ratings, matchmaking and public lobby, chat, four-human mode, computer opponent, tournaments, native apps, dark theme, languages other than English. A game record stores seats rather than accounts so accounts can be added later.

4.3 SUPPORTED BROWSERS

Latest two versions of Chrome, Edge, Firefox and Safari (including iOS Safari 16 and later). Touch and pointer are both first-class.

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5. Interaction design

5.1 THE ATTENTION PROBLEM

The product's real subject is divided attention. Every screen element serves one of four questions: *what is the time on each board, which board needs me, what is in my reserve, what is my opponent about to receive.*

5.2 LAYOUT, VIEWPORT WIDTH ≥ 1100 PX

Two boards side by side, A left and B right, joined by a **spine** `44 px` wide.

- **Strips:** each board has a top strip (the opponent seat) and a bottom strip (your seat). A strip contains, left to right: board badge (letter on the board's accent colour), name, seat colour, the seat's reserve, and the clock.
- **Orientation:** your pieces are at the bottom on both boards. For Team White-A that means Board A is shown with White at the bottom and Board B with Black at the bottom (flipped). For Team Black-A it is the reverse. This matches how bughouse is played physically.
- **Board identity:** A and B differ by accent colour, by the letter badge, and by fixed position. Colour is never the only cue.
- **Your move marker:** the seat that has the move on a board has its clock on a dark ground (`--clock-run-bg`). Opposite seats' clocks stay on the page ground. No other marker is used for "your turn".
- **Check and pending marker:** the king's square carries a 3 px ring in the alert colour, and the status line under the boards names the board and state in words (e.g. "Board B: check, Black to move" or "Board A: pending, needs a blocking piece").
- **Reserves:** pieces in a row with a count badge for counts above one. The opponent's reserve is shown at 80% size (24 px against 30 px) but is always visible, because it is tactical information.
- **Spine:** when a capture transfers a piece, the piece travels from the capturing board across the spine into the partner reserve in 420 ms (`--t-transfer`). It warns the opponent that you gained a piece. Under `prefers-reduced-motion` there is no travel: the reserve updates with a brief highlight.
- **Move list:** a single merged, timestamped list in a collapsible panel to the right, filterable by board. Closed by default during play.
- **Status line:** under the boards, one line per board in plain words (§7.6).

5.3 LAYOUT, SMALLER VIEWPORTS

- **600 to 1099 px wide:** boards stay side by side; strips and the spine shrink; reserves go to a single row.
- **Under 600 px, portrait:** boards are **stacked** (A above B) with the spine horizontal between them. Board size = `min(100vw - 16 px, (100dvh - 40 px header - 4 × 44 px strips - 8 px spine - safe-area) / 2)` . Strips are 44 px with a 28 px clock. On a 390 × 844 phone with a 34 px bottom safe-area inset this is about 293 px, or **36 px squares** (about 310 px with no inset); the naive stacked layout would be 1,111 px tall, so the formula is required.
- **Minimum square size is 34 px.** If the formula gives less (for example 360 × 640), switch to **focus mode**: one board at full width and the other as a 22 px-square mini board, both with full clocks and reserves visible; tapping the mini board swaps which one is large. Both remain fully playable.
- **Phone landscape:** boards side by side, sized by height with the same 34 px floor.
- Drag and tap targets are never smaller than the squares. Drops: drag from the reserve, or tap a reserve piece and then a square.

5.4 INPUT METHODS

- **Pointer:** drag-and-drop and click-click for both moves and drops. Legal targets show as small dots.

- **Premove:** while it is not your turn on a board, you may queue **one** action there (a move, or a drop of a piece currently in the reserve). When your turn on that board begins, the client sends the queued action at once, and the server validates it normally. If rejected, the premove is dropped silently with a brief mark on the square. Cancel with Esc, right-click, or clicking the origin square.
- **Keyboard command line:** open with `/` or Enter. Grammar: `<board> <notation>` where board is `a` or `b` (case-insensitive) and notation is the algebraic notation of §3.10, e.g. `a e4`, `b N@f3`, `a 0-0`, `b e8=Q`. The board prefix may be omitted when the notation is legal on exactly one board where you hold the move; if it is legal on both, the line shows "legal on A and B, add a or b". The line autocompletes from the legal-action list. Enter submits; Esc closes.
- **Keyboard navigation:** `Tab` cycles regions (board A, board B, reserve, command line, controls). `1` and `2` jump to board A and B. Arrow keys move the square cursor; Enter selects and then places; Esc cancels.
- **Destructive controls:** Resign and Offer draw each need a second confirming click within 5 s.

5.5 FEEDBACK

- **Sound** is synthesized with the Web Audio API, so no audio files exist and none need licensing. Cues: move, capture, drop, check, low-time (once, when your running clock crosses the low-time threshold), game end. Default on at 30% gain, with a per-user mute remembered locally.
- Last-action highlights per board (`--last-move`).
- A status line per board (§5.2).

5.6 ACCESSIBILITY

- Everything possible with a pointer is possible with the keyboard alone (§5.4).
- An ARIA live region announces each action with its board and seat, rate-limited to one announcement per second per board; low time and results are announced immediately.
- Text contrast ≥ 4.5:1, UI components ≥ 3:1 (the measured table is in §7.3).
- `prefers-reduced-motion` is honoured. There is no dark mode in v1, so `prefers-color-scheme` is deliberately not used.
- Piece identity never depends on colour alone; a promoted piece shows a small marker in addition to its normal glyph.
- Clock numerals have text alternatives through the live region and `aria-label`s on the clock strips.

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6. Technical architecture

6.1 STACK (DECIDED)

- TypeScript throughout. npm workspaces: `packages/engine`, `packages/server`, `packages/web`.
- **Client: Vite + Svelte 5 SPA.** Chosen for small bundles and a reactive model that suits four frequently updating clocks. (React is not used.)
- **Server and hosting: Cloudflare Workers + Durable Objects**, one object per game. A Durable Object serializes all events (deterministic ordering across the two boards), holds the WebSockets, and uses alarms for flag timeouts. The client is served from the same platform. Vercel is not used because it cannot hold long-lived WebSockets.
- **Persistence: Cloudflare D1** for finished games; retention 90 days.
- **Contingency:** if Durable Objects prove unworkable (alarm precision or WebSocket limits found in M3 testing), switch to Node + `ws` on Fly.io with the same `engine` package. This is a fallback with a defined trigger, not a co-equal option.
- Test tooling: Vitest, Playwright, axe.

6.2 AUTHORITATIVE SERVER AND CLOCKS

- The server alone decides legality, clocks and results. Clients render and request.
- A clock is stored as `(remainingMs at lastEventTs, running)`. Clients interpolate from a server-synchronised offset and never decide a flag.
- A flag is detected by a Durable Object alarm set to the earliest running deadline and rescheduled after every action.

- **Lag credit:** per accepted action, the server credits the mover `min(RTT/2, 150 ms)`, where RTT is the median of the player's last five ping measurements (pings every 5 s). The credit is added back to the mover's bank, and never more than the time elapsed on that move.
- All events are serialised in the object, which makes cross-board effects (reserves) consistent. An attempt to drop a piece in the same instant as the capture that supplies it is decided by arrival order.
- Event-order and tie rules: §3.6.1.

6.3 ENGINE (BUILT; M1 DONE)

Source of truth: `packages/engine`. Shape as implemented:

- `Game = { pos: {A, B}, reserve: {A: {w, b}, B: {w, b}}, opts }` where `reserve[board][color]` is the reserve of that seat.
- `legal(game, board)`, `apply(game, board, action) → { game, captured, transfer }`, `status(game, board)` returning `ok | check | pending | mate | stuck`, and `result(game)` for mate on either board. Clocks, flags, resign and draws live in the server, not the engine.
- Verified in `npm test`: perft and 300 random playouts against `chessops` Crazyhouse (single-board legality), 35 named bughouse scenarios, and 400 random two-board games asserting that the piece census is conserved (32 pieces of each colour; promoted pieces counted as pawns) and that every legal action applies. Nine deliberate rule breaks each made the suite fail.

6.4 PROTOCOL

Messages are JSON, schema-validated on both sides, each server message carrying a monotonically increasing `seq`.

- Client → server: `join{name, seatToken?}`, `setup{timeControl, team}` (host only, before Ready), `ready{value}`, `action{board, action}` (move: `from,to,promo?`; drop: `piece,to`), `resign`, `drawOffer`, `drawAccept`, `drawDecline`, `rematch`, `ping{t}`.
- Server → client: `snapshot{full state, seq}`, `action{board, action, seq, serverTs, clocks}`, `reject{reason}`, `ended{result, reason}`, `presence{seat, connected}`, `drawOffered`, `pong{t, serverTs}`.
- Gap in `seq` → the client asks for a snapshot. Reconnect uses the same path.

6.5 PRESENCE AND ABANDONMENT

- A human is **disconnected** after their last WebSocket closes. Clocks keep running.
- After **60 s** of continuous disconnection, the connected opponent may press "Claim win"; the server verifies the disconnection before awarding it. If the disconnected human reconnects first, play continues.
- If **both** humans are disconnected for **5 minutes**, the game is voided: no result.

6.6 IDENTITY AND SECURITY

- Table ids are 8 characters from a 32-symbol alphabet without look-alike characters (about 40 bits), unguessable and unlisted.
- A seat is claimed with a random 128-bit seat token held in local storage and presented on connect. Spectators have no token and cannot send actions.
- Rate limits per connection (30 messages per second); the server rejects any message failing schema validation.
- No third-party trackers or analytics in v1.

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7. Design: "Club table" (decided 2026-10-07)

Sources of truth: DESIGN.md for decisions and design/tokens.css for values. A rendered reference of the game screen is design/preview.html. The two alternates (Ledger, Reading room) were rejected and remain in `gate1/` only for the record.

7.1 WHAT THE REFERENCE ARTICLE SAYS, AND HOW IT IS APPLIED

Reference: Leah Finch, **"How to Avoid AI Slop in Web Design: A Beaver Builder AI Masterclass with Paul Lacey"**, Beaver Builder, 11 September 2026, `https://www.wpbeaverbuilder.com/avoid-ai-slop-in-web-design/`, read in full on 2026-10-06. The points used:

- **AI fills every unspecified decision with its defaults** ("If you don't decide, the AI decides for you"). Cure: decide every design variable explicitly (DESIGN.md) and keep approval gates (§12).
- **Named hallmarks:** repeated eyebrow labels and decorative monospace captions above headings; headlines with italicised words, gradients and dramatic three-part slogans; glassmorphism, translucent cards, floating gradients, boxes nested inside boxes. These are turned into the rejection checklist (§10).
- **The techniques are not banned, only unjustified.** Our test: each visual device must say what it does for a bughouse player, or be cut.
- **Too few rules produced slop, too many produced safe and uninspired work.** Hence a short token set (the tokens file is about 50 lines) plus one explicit checklist, not a long style guide.
- **Approval gates** on style tiles, typography, colour and then the main screen before building the rest, which §12 adopts.

7.2 DEFAULT REFLEX AGAINST DECISION

Default reflex	Decision
Dark navy/black UI with a neon accent	Warm matte light "club table"; the boards are the only saturated objects
Inter, JetBrains Mono, Space Grotesk, Playfair (and what earlier projects used)	**Atkinson Hyperlegible Next** for UI and clocks, **Zilla Slab** for titles only. No monospace
Hero with gradient, tagline and three feature cards	No hero. The first screen is the table form beside a live engine-driven demo of a capture crossing between boards
Rounded cards, soft shadows, glass	Flat panels divided by 1 px rules; 2 px radius on controls; boards and panels square; no shadows
Decorative icons and emoji	The chess pieces and a minimal functional icon set (resign, draw, mute, move list); no emoji
Pulsing red low-time alarm	The clock ground becomes solid alert colour and the weight increases to 800; plus one sound cue; no animation loop
Centered, symmetric layout	Board-first layout dictated by the two-board geometry and the spine

7.3 PALETTE (FINAL; CONTRAST MEASURED 2026-10-07)

Token	Value
Page / panel / rule	`#ECE6D8` / `#E3DBC9` / `#CFC5AE`
Ink / secondary ink	`#1E1B16` / `#5A5444`
Board A: light / dark square / accent	`#F0E3C4` / `#BA7C52` / `#B5482F` (clay)
Board B: light / dark square / accent	`#DDE7E2` / `#6D9696` / `#1F5A63` (deep teal)
Alert (check, pending, low time)	`#C2410C`
Running clock	background `#1E1B16`, text `#ECE6D8`

Measured contrast ratios: ink on page 13.8; secondary ink on page 6.1 and on panel 5.5; white on clay badge 5.3; white on teal badge 7.8; running clock text 13.8; white on alert 5.2; clay against page 4.3; teal against page 6.3; black pieces on A dark 5.5, on B dark 5.8; white pieces on A dark 3.5, on B dark 3.3 (light-piece outlines carry the rest). The first draft's A dark square (`#C58A5F`) gave 2.9 for white pieces and was darkened to `#BA7C52`.

7.4 TYPOGRAPHY (FINAL)

- **UI, body and clocks: Atkinson Hyperlegible Next** (variable, 400 to 800). Designed by the Braille Institute to keep similar characters apart (0/8, 1/l), which matters for a clock read at a glance. Tabular numerals were verified (equal width for `1111` and `0000`).
- **Titles, wordmark and result line only: Zilla Slab** (600 and 700). A slab with a scoresheet feel; not a typical AI default.
- Scale: clock 36 px (28 px under 640 px), title 28 px, body 15 px, small 13 px (the minimum text size).
- Self-hosted latin subsets in `design/fonts/` with `font-display: swap`; the clock font is preloaded. Both families are SIL OFL 1.1, confirmed from the google/fonts repository; the licence texts are saved beside the fonts and recorded in `THIRD_PARTY.md`.
- Replaced: Schibsted Grotesk and Young Serif (the first draft). Evaluated and rejected: Archivo, Familjen Grotesk, Hanken Grotesk, Literata, Gloock, Source Serif 4.

7.5 COMPONENTS, EACH WITH A REASON

- **Clock strip:** the largest numerals on screen, next to the board edge. No skeuomorphic clock faces.
- **Reserve row:** pieces with count badges; the opponent's slightly smaller.
- **Spine:** the capture-transfer travel (§5.2).
- **Result screen:** one plain sentence of who won and why, both final boards, the action list. No confetti.

7.6 COPY

Dry, concrete, second person, no slogans, no exclamation marks, no "Not X. It's Y." constructions. Examples: "Set a table", "Waiting for your opponent", "Board B: Black to move, 0:42", "Board A: pending, needs a blocking piece", "Black wins on Board B: checkmate".

7.7 PIECES

Cburnett SVGs (§2 U4, files in `design/pieces/`), shown at 92% of the square. Legibility is verified at 36 px squares in the preview.

---

8. Information architecture

Route	Purpose
`/`	Set a table (time control, team) and the demo boards
`/t/:id`	Table → game (single page with states: setup, countdown, playing, result); forwards to `/g/:id` once finished
`/g/:id`	Finished-game replay
`/watch/:id`	Spectator view
`/rules`	Full rules with interactive mini-boards

Unknown id → a plain "no such table" page with a link to `/`. No about, pricing or blog pages.

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9. Quality bars

9.1 RULES TESTS (EXIST; MUST STAY GREEN)

- Perft comparison against `chessops` Crazyhouse to depth 3 on 11 positions (depth 4 on the initial position), plus 300 random games of 60 plies comparing move counts at every ply.
- Named scenarios for each rule in §3 (transfer colour and seat, counterfeits, drop legality including in check, pawn drop ranks, double step, en passant on dropped pawns, promotion choices, check/mate/pending/stuck, dropped-rook castling on both flag settings).

- Property tests: piece-census conservation over 400 random two-board games; every legal action applies; reserves never negative; two kings per board.
- Mutation discipline: every key rule has a deliberate-break check that must make the suite fail. Server-layer rules (clock order, ties, increments, lag credit, abort, abandonment) must get the same treatment in M3.

9.2 SERVER TESTS

Clock accuracy under simulated latency; event ordering including §3.6.1 cases (action at the deadline, simultaneous flags for same and different teams); alarm flags; reconnect and snapshot resync; abandonment claim; abort; message-validation fuzzing; rate limiting.

9.3 END-TO-END (PLAYWRIGHT, TWO BROWSER CONTEXTS)

Scripted full game: create → join → ready → opening moves on both boards → capture → transfer → drop → mate. A second game ending on a flag on Board B. A pending-mate game resolved by a drop. Reconnect mid-game. Spectator is read-only. Replay scrub. Visual snapshots at 1280, 768 and 390 widths. axe audit. A keyboard-only playthrough.

9.4 PERFORMANCE AND LICENSING

- Measured with Lighthouse using its mobile preset on the production build: **initial-route JavaScript ≤ 150 KB gzipped** (fonts and piece SVGs excluded), **LCP < 1.5 s**, and 60 fps during drags and the spine animation in the browser performance trace.
- `THIRD_PARTY.md` records licence, author and source URL for the pieces, the fonts and every runtime dependency.

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10. Anti-slop checklist

Who enforces: the builder runs it before every gate; the `/design-review` skill is run before Gates 3 and 4. A change is rejected if any of these is true:

- an eyebrow or kicker label, or decorative monospace caption, above a heading
- italic-accent word headlines, gradient text, or three-beat slogans
- glassmorphism, translucent panels, glows, floating gradient blobs
- a container nested inside another container without a functional reason
- three identical feature cards
- emoji or stock icons used as decoration
- a colour, shadow, radius or animation that cannot be justified in one sentence for a player
- any value (font, size, space, radius, easing) not present in `design/tokens.css`

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11. Milestones and exit criteria

M	Deliverable	Exit criterion	Status
M0	`DESIGN.md`, `design/tokens.css`, fonts, pieces, `THIRD_PARTY.md`	Direction approved, contrast measured, licences recorded	Done 2026-10-07: direction, tokens, fonts, pieces, licences verified
M1	`engine` package	§9.1 green	**Done 2026-10-07**
M2	Local two-seat prototype (no network)	One person plays a full game against themself on both boards with all rules and four local clocks, in the Club table design	Next
M3	Server rooms, clocks, protocol	Two browsers complete a game; §9.2 green	
M4	Full UI, responsive, keyboard entry, premoves, spine	§9.3 passes at three widths	
M5	Replay, spectators, export, rematch, /rules	All of §4.1 shipped	
M6	Hardening	§9 bars met	

12. Approval gates

- **Gate 0:** brief. Reviewed and revised at the user's direction; final export requested 2026-10-07.
- **Gate 1:** visual direction. **Closed 2026-10-07: Club table.**
- **Gate 2:** rules engine green. **Closed 2026-10-07.**
- **Gate 3:** the user plays one full game on the M2 prototype, then the design is reviewed against §10 before polish begins.
- **Gate 4:** pre-ship review: §9 bars met, §10 checklist clean, `THIRD_PARTY.md` complete.

13. Risks

Risk	Mitigation
Pending/mate logic subtly wrong	Isolated in the engine; perft oracle; scenarios; mutation checks (done)
Latency decides flags unfairly	Server-authoritative clocks, lag credit capped at 150 ms, tested under injected latency
Phones too cramped for two boards	Computed board size, 34 px square floor, defined focus mode (§5.3); real-device test in M4
Durable Object timing or WebSocket limits	Fallback to Node + `ws` on Fly.io with the same engine, triggered if M3 alarm tests miss ±50 ms
Piece set looks generic	Accepted for v1 for licence certainty; custom set is a post-v1 upgrade
Players confuse this format with four-player bughouse	Lobby copy and /rules state plainly that each human plays both boards

14. Decisions on previously open points (final; reversible on request)

1. U1 dropped-rook castling: allowed (flag). U2 promotion: Q/R/B/N. U3 repetition/50-move: none. U4 pieces: Cburnett.
2. Name: "Tandem" is the working title; renaming is a one-line config change.
3. Accounts and ratings: out of v1.
4. Increments: optional, default none.

5. Guest names only; 1 to 20 characters; no profanity filter in v1.
6. Retention: 90 days for finished games; unjoined tables expire after 30 minutes.
7. Dark theme: not in v1.
8. Hosting: Cloudflare Durable Objects + D1; Node + `ws` on Fly.io is a defined fallback only.

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15. References

- Reference article: Leah Finch, “How to Avoid AI Slop in Web Design: A Beaver Builder AI Masterclass with Paul Lacey”, 11 Sep 2026. `https://www.wpbeaverbuilder.com/avoid-ai-slop-in-web-design/`
- Bughouse rules: Wikipedia, “Bughouse chess” `https://en.wikipedia.org/wiki/Bughouse_chess`; Laws of Bughouse (wccc2007) `https://www.chessfed.gr/old_site/wccc2007/files/bughouse_chess_complete_rules.pdf` and `https://www.chessfed.gr/old_site/wccc2007/files/bughouse_chess_main_rules.pdf`; complete ruleset `http://www.pion.ch/Bug/ruleseng.html`; chess.com Help Center `https://support.chess.com/article/349-bughouse`; wikibooks `https://en.wikibooks.org/wiki/Chess/Variants/Bug_House`.
- Castling with dropped rooks (crazyhouse, contrasting): `https://lichess.org/forum/lichess-feedback/crazyhouse-castling-bug`, `https://talkchess.com/forum3/viewtopic.php?t=72124`.
- Single-board legality oracle: `chessops` (lichess), `https://github.com/niklasf/chessops`.
- Pieces: Cburnett, `https://commons.wikimedia.org/wiki/File:Chess_klt45.svg` (licences listed on that page).
- Fonts: Atkinson Hyperlegible Next and Zilla Slab via Google Fonts, `https://fonts.google.com/specimen/Atkinson+Hyperlegible+Next`, `https://fonts.google.com/specimen/Zilla+Slab`.

Changelog

v2.1 (2026-10-07), final pass. Fixed: king listed as a possible checking piece (§3.4); phone board-size figure now states its safe-area assumption (§5.3); Gate 0 no longer claims an approval that was not given; piece licence changed from BSD (verified on one file) to CC BY-SA 3.0 (verified on all twelve); font licences verified against the google/fonts repository and the OFL texts added; heading levels corrected in the compiled master.

v2.0 (2026-10-07). User chose Club table and asked for replacement fonts, a citation check, and an ambiguity audit. Changes:

- Fonts replaced (Schibsted Grotesk and Young Serif out; Atkinson Hyperlegible Next and Zilla Slab in), tested by rendering and a tabular-width measurement. In my preview pane the earlier style tiles never loaded their web fonts (external stylesheets are blocked there), so my own check of them showed fallback type. Corrected by self-hosting the fonts.
- Palette contrast measured; Board A’s dark square changed to fix a failing 2.9 ratio.
- Reference article: full citation with author, date, URL and which claims were used. All sources and licences now listed in §15.
- Ambiguities resolved: mate condition completed (pawn, knight, double, no-gap); capture availability (“next time that seat has the move”, not “very next action”); event ordering and ties defined; draw offer rules; clock start, abort and increment semantics; time display and low-time threshold; command-line grammar; premove semantics; keyboard navigation; phone layout numbers and focus mode; “your move” marker; table, seat and spectator rules; id format; lag credit constants; abandonment; export format; browser support; hosting and framework decisions made rather than listed as options; `prefers-color-scheme` contradiction removed; engine status values corrected to five; glossary added; “Alternatives” no longer left open.
- Engine milestone and test counts updated to match what is built.

Part 2. Design system

Task-zero design record. Values live in design/tokens.css; rendered reference in design/preview.html. Nothing in the UI may use a colour, size, space, radius or easing that is not in the tokens. Deviations need explicit approval. Decided 2026-10-07; the earlier alternates are archived in `gate1/`.

Principle

Every visual device must say what it does for a bughouse player, or be cut. The player's real problem is divided attention across two boards; the design's job is to make four clocks, four reserves and two positions readable at a glance.

Surfaces

Warm matte paper (`--page`), one slightly darker panel (`--panel`), 1 px rules (`--rule`). No shadows, no gradients, no translucency. Containers do not nest. Boards and panels have square corners; controls, badges and clocks use `--radius: 2px`.

Colour

- **Board identity.** A is clay (`--a-accent`, squares `--a-sq-*`); B is deep teal (`--b-accent`, squares `--b-sq-*`). Each board also has a letter badge and a fixed position (A left, B right). Colour is never the only cue.
- **State.** `--alert` appears only for check, pending and low time. `--clock-run-bg` marks the seat whose clock is running.
- Contrast measured in BRIEF §7.3.

Type

- **Atkinson Hyperlegible Next** for UI, body and clocks (tabular figures verified).
- **Zilla Slab** for the wordmark, page titles and the result line only.
- Scale: clock 36/28 px, title 28, body 15, small 13 (minimum). No monospace. No italics for emphasis.

Components

- **Clock strip:** badge, name, seat colour, reserve, clock. The clock is the largest text on screen. Running: dark ground. Low: solid alert ground, weight 800. No pulsing.
- **Reserve:** pieces with count badges; the opponent's at 24 px, yours at 30 px.
- **Spine (44 px):** capture-transfer travel in 420 ms; static highlight under reduced motion.
- **Status line:** plain words per board.
- **Result:** one sentence, both final boards, action list. No confetti.

Motion

120 ms for piece movement, 420 ms for the transfer, one easing curve (`--ease-out`). Zero under `prefers-reduced-motion`.

Copy

Dry, concrete, second person. No slogans, exclamation marks, or "Not X. It's Y." constructions.

Rejection checklist (reference: wpbeaverbuilder.com/avoid-ai-slop-in-web-design)

Reject any change containing: eyebrow labels or decorative monospace captions; italic-accent headlines, gradient text, three-beat slogans; glassmorphism, translucent panels, glows, floating gradients; nested containers without a function; three identical feature cards; decorative emoji or stock icons; values not in the tokens.

Pieces and assets

Cburnett SVG set (`design/pieces/`), CC BY-SA 3.0, shown at 92% of the square. Fonts self-hosted in `design/fonts/` with their OFL texts. All licences are recorded and verified in `THIRD_PARTY.md`.

Appendices

Appendix A. Design tokens (verbatim from `design/tokens.css`)

```
/* Tandem design tokens. Source of truth: DESIGN.md. Do not introduce values that are not here. */
@import url("./fonts/fonts.css");

:root {
  /* Surfaces and ink (light only in v1) */
  --page:   #ECE6D8;
  --panel:  #E3DBC9;
  --rule:   #CFC5AE;
  --ink:    #1E1B16;
  --ink-2:  #5A5444;

  /* Board identity. Each board = colour + letter badge + fixed position (A left, B right). */
  --a-accent: #B5482F;  --a-sq-light: #F0E3C4;  --a-sq-dark: #BA7C52;
  --b-accent: #1F5A63;  --b-sq-light: #DDE7E2;  --b-sq-dark: #6D9696;
  --on-accent: #FFFFFF;

  /* State colours. Used only for state, never decoration. */
  --alert:        #C2410C;   /* check, pending, low time */
  --on-alert:     #FFFFFF;
  --clock-run-bg: #1E1B16;   /* the side whose clock is running */
  --clock-run-ink:#ECE6D8;
  --last-move:    rgba(232, 197, 71, .38);

  /* Type */
  --font-ui:    "Atkinson Hyperlegible Next", system-ui, sans-serif;  /* UI, body, clocks */
  --font-title: "Zilla Slab", Georgia, serif;                          /* wordmark, page titles, result line
only */
  --fs-clock:   2.25rem;   /* 36px; 1.75rem (28px) under 640px */
  --fs-title:   1.75rem;
  --fs-body:    0.9375rem; /* 15px */
  --fs-small:   0.8125rem; /* 13px; minimum text size */
  --fw-clock:   600;
  --fw-clock-low: 800;

  /* Space (4px base) and shape */
  --s-1: 4px; --s-2: 8px; --s-3: 12px; --s-4: 16px; --s-5: 24px; --s-6: 32px; --s-7: 48px;
  --radius: 2px;          /* controls, badges, clocks. Boards and panels are square. */
  --spine-w: 44px;

  /* Motion */
  --t-move: 120ms;
  --t-transfer: 420ms;
  --ease-out: cubic-bezier(.2, .7, .2, 1);
}

@media (prefers-reduced-motion: reduce) {
  :root { --t-move: 0ms; --t-transfer: 0ms; }
}

html { background: var(--page); color: var(--ink); font: var(--fs-body)/1.45 var(--font-ui); }
.clock { font-family: var(--font-ui); font-weight: var(--fw-clock); font-variant-numeric: tabular-nums; font-
size: var(--fs-clock); line-height: 1; letter-spacing: -.01em; }
@media (max-width: 640px) { :root { --fs-clock: 1.75rem; } }
```

Appendix B. Rules engine

Pure TypeScript bughouse rules. No DOM, no network, no clocks (clocks live in the server, brief §6.2).

- `board.ts`: one board. Moves, drops, promotion (flagged), en passant, castling (incl. dropped-rook option), check, `boardStatus` = `ok | check | pending | mate | stuck`.
- `game.ts`: two boards + four reserves. `apply` validates, consumes drops, and transfers captures to the partner seat (colour preserved, promoted pieces become pawns). `result` ends the game on mate on either board.
- `fen.ts`: crazyhouse-style FEN with `[pockets]` and `~` promoted marks, for tests and fixtures.

Seats: A-w and B-b are one human's seats, A-b and B-w the other's.

VERIFICATION (`NPM TEST`)

- `oracle.test.ts`: perft and 300 random playouts against lichess `chessops` Crazyhouse (single-board legality). `droppedRookCastling: false` there because lichess forbids it.
- `bughouse.test.ts`: named scenarios for every rule in BRIEF §3 (transfer, counterfeits, drops, pending vs mate, no stalemate, dropped-rook castling).
- `property.test.ts`: 400 random two-board games asserting piece census conservation (32 per colour, promoted counted as pawns) and that every legal action applies.
- Mutation checked: nine deliberate rule breaks (pawn drops on rank 1/8, pending=mate, promoted capture keeps type, wrong transfer colour, dropped-rook flag ignored, no double step, no en passant, drops while in check, stalemate=mate) each make the suite fail.

Not here yet: clock layer, flags, draw/resign (server, M3).

Test run for this compilation: 3 files, 49 tests, all passing.

Appendix C. Third-party licences

All licences below were checked on 2026-10-07 against the sources named.

Asset	Author / source	Licence	Where used	Ho
Chess pieces (12 SVGs, Cburnett set)	Cburnett. Files `Chess_{k,q,r,b,n,p}{l,d}t45.svg` on Wikimedia Commons, e.g. `https://commons.wikimedia.org/wiki/File:Chess_klt45.svg`	**CC BY-SA 3.0** (attribution; share-alike applies to modified versions of the artwork). The author also offers GFDL 1.2+, GPL 2+ and BSD 3-clause on at least `Chess_klt45.svg`, but only CC BY-SA 3.0 was confirmed on all twelve files, so that is the one we rely on	`design/pieces/`	Wi `e` rep
Atkinson Hyperlegible Next	The Atkinson Hyperlegible Next Project Authors (Braille Institute, Applied Design Works, Elliott Scott, Megan Eiswerth, Letters From Sweden), `https://github.com/googlefonts/atkinson-hyperlegible-next`	**SIL Open Font License 1.1.** Copyright 2020-2024 The Atkinson Hyperlegible Next Project Authors	`design/fonts/`	`OF` (`l` go as `At`
Zilla Slab	The Mozilla Foundation; designer Typotheque	**SIL Open Font License 1.1.** Copyright 2017 The Mozilla Foundation	`design/fonts/`	`OF` (`l` go as `Zi`
chessops	Niklas Fiekas, `https://github.com/niklasf/chessops`	GPL-3.0-or-later	Dev dependency, used only as a test oracle; not shipped	`pa` Be sit lic

OBLIGATIONS

- **Pieces:** show the attribution below on /rules and keep the SVGs unmodified. If a piece is ever edited, the edited artwork must be released under CC BY-SA 3.0.
- **Fonts:** the OFL text ships alongside the font files (done). The fonts may be bundled and served but not sold on their own.

Attribution text for pieces (shown on /rules): “Chess pieces by Cburnett, licensed CC BY-SA 3.0 (`https://creativecommons.org/licenses/by-sa/3.0/`).”

Still unchecked: runtime npm dependencies, because none exist yet (Svelte, Vite and others arrive in M2 to M3). Record each when it is added.

Appendix D. File inventory

Path	What it is
`MASTER_DESIGN.md`	This document, compiled by `tools/build_master.py`
`MASTER_DESIGN.pdf`	PDF export of this document
`BRIEF.md`	Build brief: product, rules, interaction, architecture, quality bars, milestones
`DESIGN.md`	Club table design record and rejection checklist
`THIRD_PARTY.md`	Licences for pieces, fonts and dev dependencies
`design/tokens.css`	Every design value (colour, type, space, motion)
`design/fonts/`	Self-hosted Atkinson Hyperlegible Next and Zilla Slab (latin subsets) and their OFL texts
`design/pieces/`	Cburnett SVG piece set (12 files)
`design/preview.html`, `design/preview.png`	Rendered reference of the game screen using the tokens
`packages/engine/`	Rules engine (TypeScript) and its tests
`tools/build_master.py`	Rebuilds this document
`gate1/`	Archived style tiles (three directions) and the font test page